Weekly Practice and Troubleshooting

Assignment 4-5: Resolving connectivity issues in a complex network

Lab Setup

The lab connects a LAN with two VLANs and a second LAN to an "ISP" router that hosts a web server. The server deliberately uses 192.168.10.2, an address that overlaps with VLAN 10 on the WAN router side, so NAT is needed on both routers to keep the two networks separate.

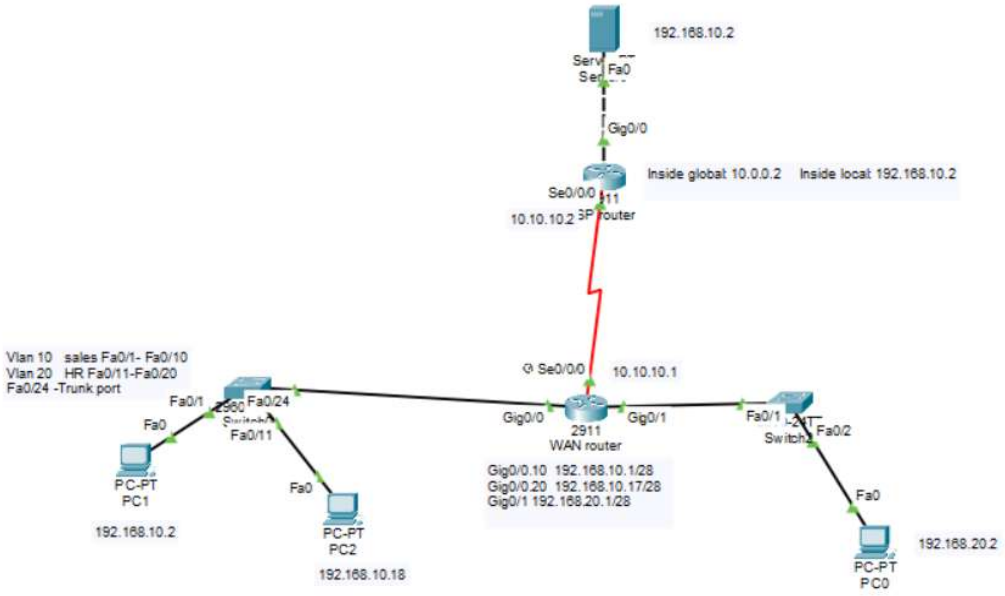


Figure 1: Network topology (the diagram labels still show the original 10.10.10.2 NAT address; the final server address is 10.0.0.2)

Device	Interface	Address	Role
WAN router	Gig0/0.10	192.168.10.1/28	VLAN 10 (Sales) gateway, NAT inside
WAN router	Gig0/0.20	192.168.10.17/28	VLAN 20 (HR) gateway, NAT inside
WAN router	Gig0/1	192.168.20.1/28	Switch2 LAN gateway, NAT inside
WAN router	Se0/0/0	10.10.10.1/30	Link to ISP, NAT outside
ISP router	Se0/0/0	10.10.10.2/30	Link to WAN router, NAT outside
ISP router	Gig0/0	192.168.10.1/24	Server LAN, NAT inside
PC1 / PC2	Fa0	192.168.10.2 / 192.168.10.18	VLAN 10 and VLAN 20 hosts
PC0	Fa0	192.168.20.2	Host on Switch2
Server	Fa0	192.168.10.2 (seen as 10.0.0.2)	Web server, static NAT on ISP router

NAT design: the WAN router uses PAT (overload) so every LAN host appears as 10.10.10.1 to the ISP side. The ISP router uses static NAT so the server appears as 10.0.0.2 to the WAN side. The WAN router reaches it with a host route.

Task 1: Use the ping command to test connectivity between two PCs on different subnets in your Packet Tracer network and document the output for both successful and failed attempts.

Method

PC1 and PC2 are in the 192.168.10.0 range (VLAN 10 and VLAN 20), and PC0 is in 192.168.20.0/28. Traffic between subnets must go through the WAN router, so each PC needs the correct default gateway. I used the Command Prompt on each PC and pinged the ISP router link (10.10.10.2) and the server (10.0.0.2).

Failed attempt

Pings from PC0 and PC1 to 10.0.0.2 timed out with 100% loss (4 sent, 0 received). In the same window, pings to 10.10.10.2 succeeded with 0% loss.

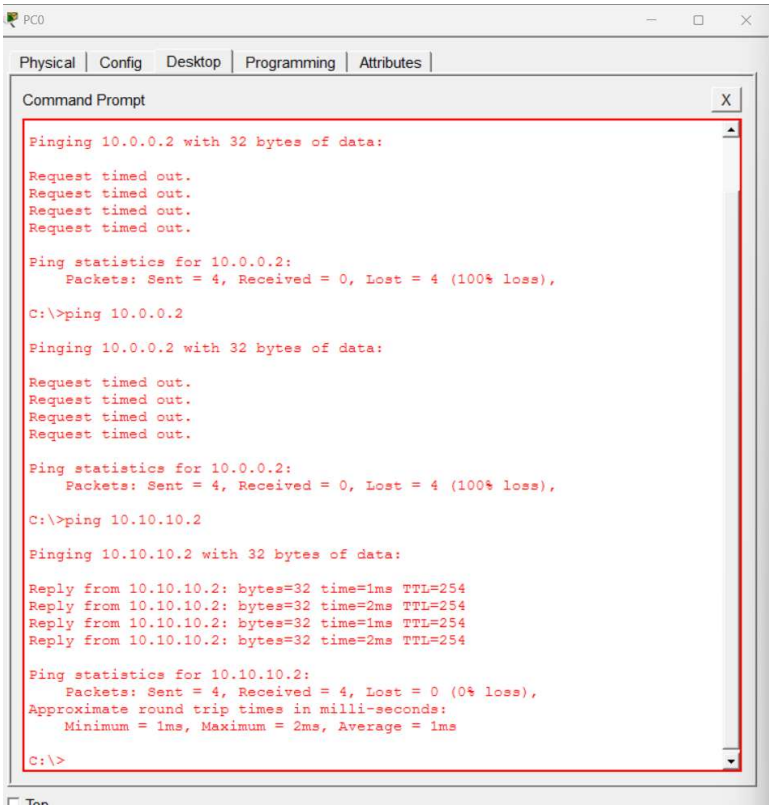


Figure 2: PC0: ping to 10.0.0.2 fails (100% loss), ping to 10.10.10.2 succeeds

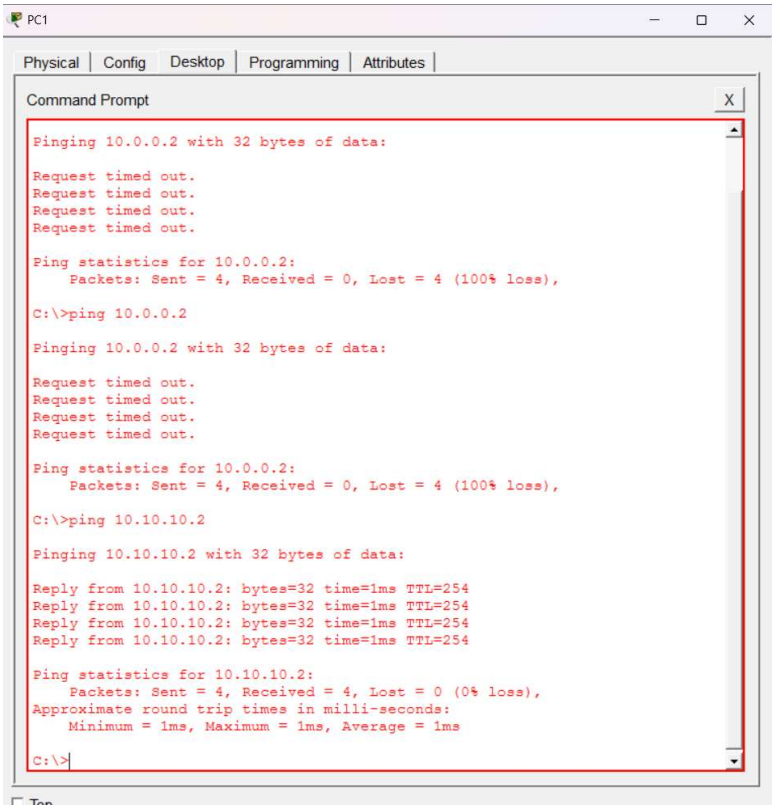


Figure 3: PC1: same result, 10.0.0.2 fails and 10.10.10.2 succeeds

What the result tells us

- The replies from 10.10.10.2 show TTL=254, which means the packet crossed one router (the WAN router). So the PC, its gateway, the VLAN trunk and the serial link all work.
- Because only 10.0.0.2 fails, the fault is beyond the ISP router: in NAT or routing on the path to the server. Task 5 shows how it was found.

Successful attempt

After the fixes in Task 5, the WAN router pinged 10.0.0.2 with 5/5 success, and its NAT table shows the LAN host 192.168.10.2 translated to 10.10.10.1 (PAT), including a TCP port 80 entry from the web request.

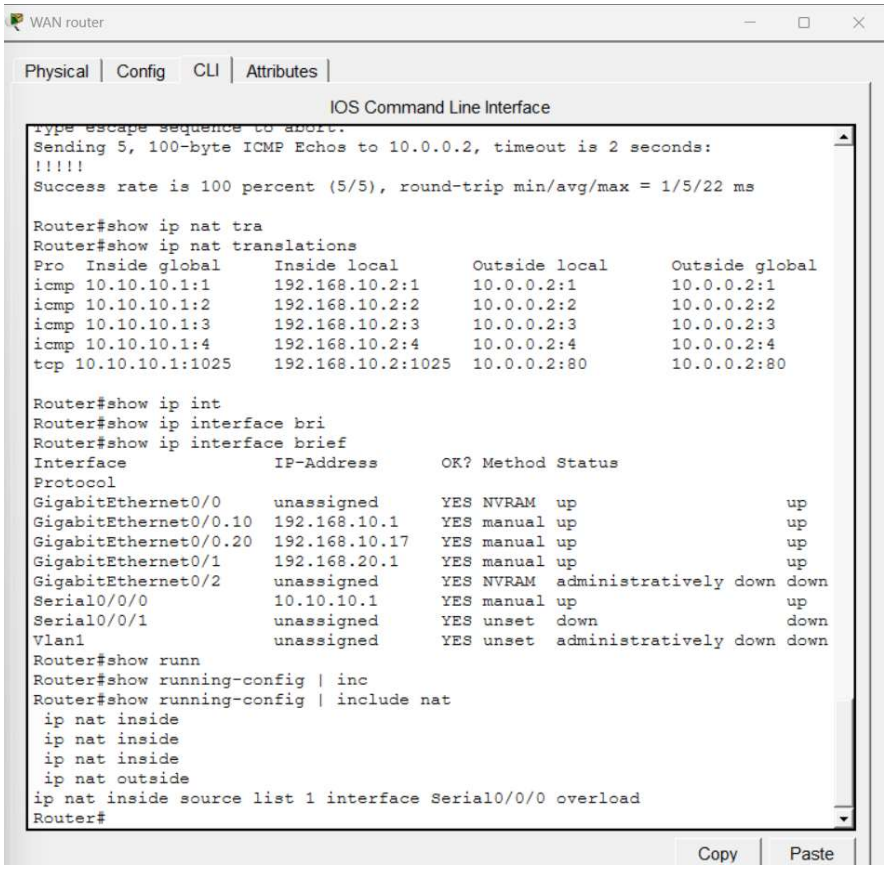
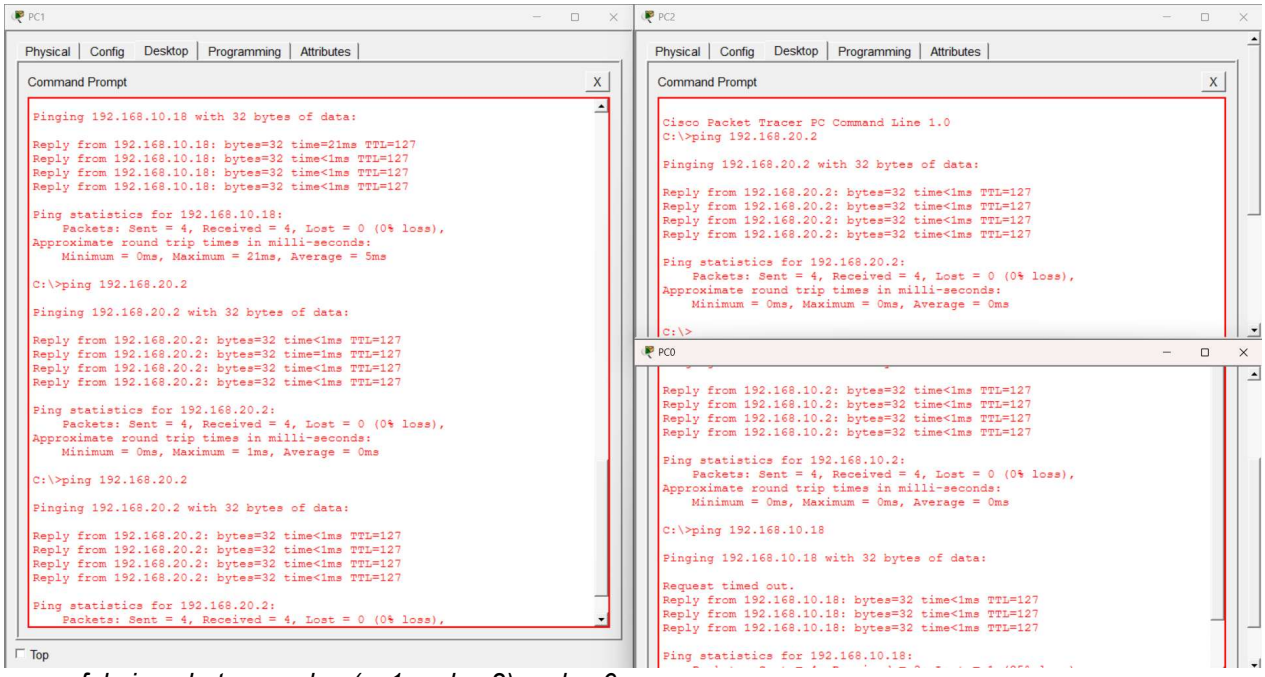


Figure 4: WAN router: 5/5 ping to 10.0.0.2 and the PAT translation table



successful pings between vlan (pc1 and pc2) and pc0

Task 2: Identify and fix an incorrect default gateway configuration on one device in your network so that it can access an external web server simulation; submit the steps you took and the command outputs before and after the fix.

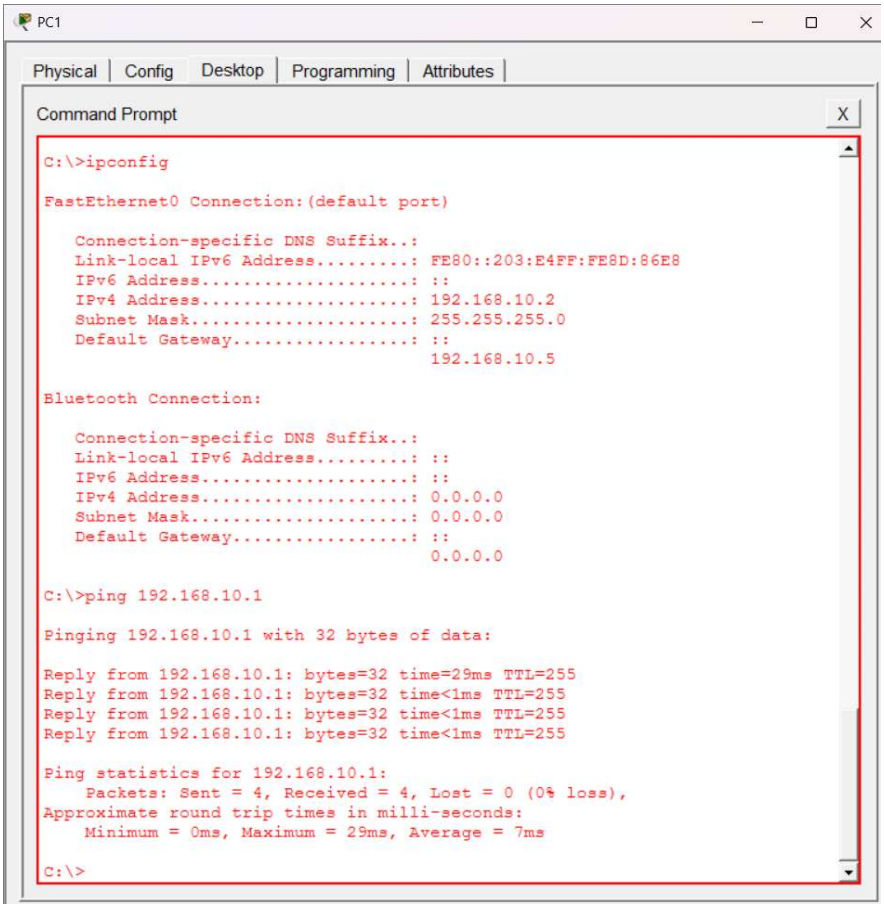
Fault introduced

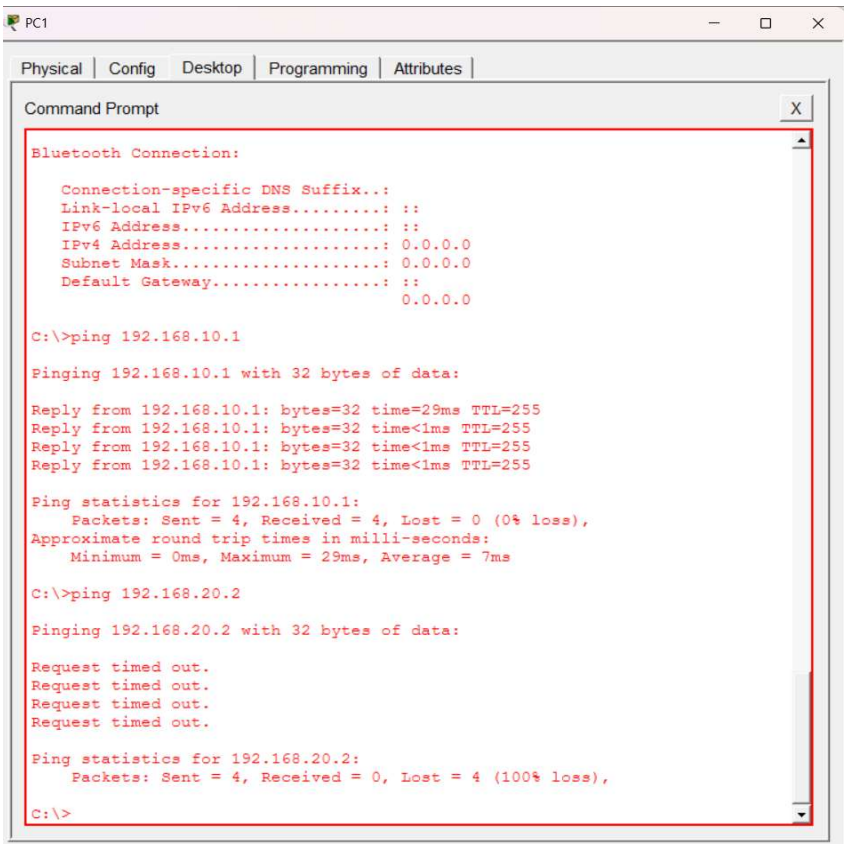
On PC1 (VLAN 10, 192.168.10.2/28) the default gateway was changed from 192.168.10.1 (the WAN router subinterface Gig0/0.10) to 192.168.10.5, an address inside the subnet that no device owns.

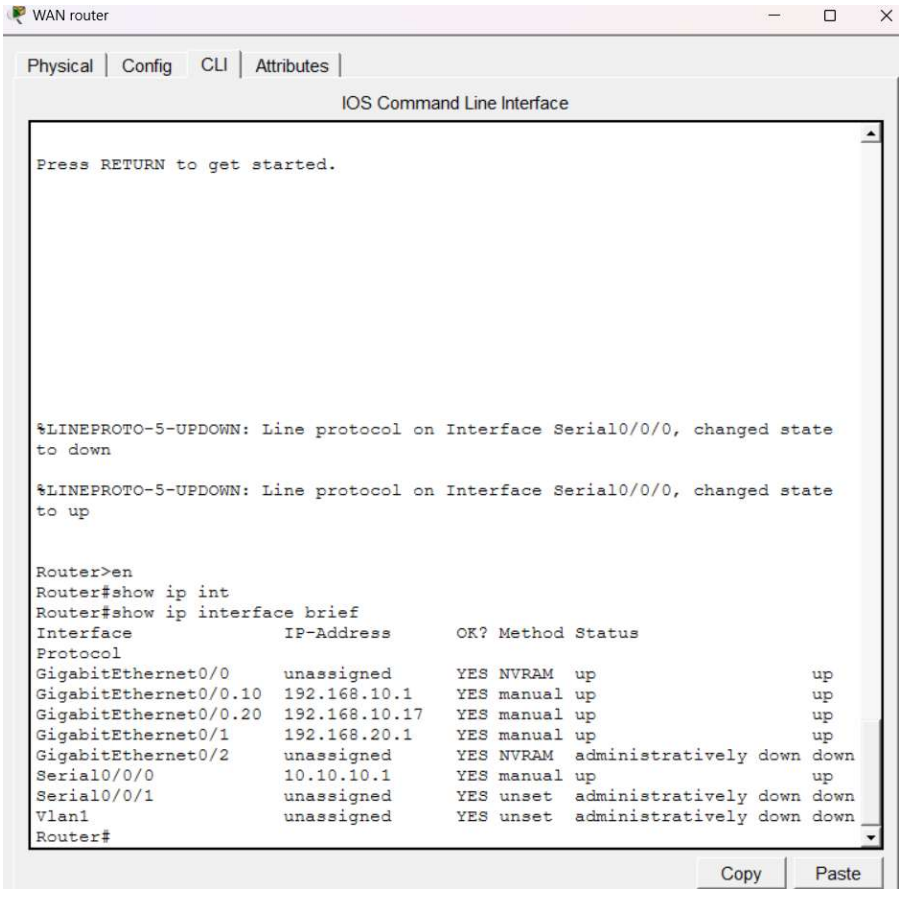
Steps taken

1. On PC1, ran ipconfig and noted the default gateway 192.168.10.5.
2. Pinged 192.168.10.1 (works, same subnet) and 10.10.10.2 and 10.0.0.2 (fail, they are off-subnet).
3. On the WAN router, ran show ip interface brief and saw that the VLAN 10 gateway is 192.168.10.1, not .5.
4. Corrected the gateway on PC1 under Desktop > IP Configuration to 192.168.10.1.
5. Re-tested: ping 10.0.0.2 and opened http://10.0.0.2 in the PC web browser.

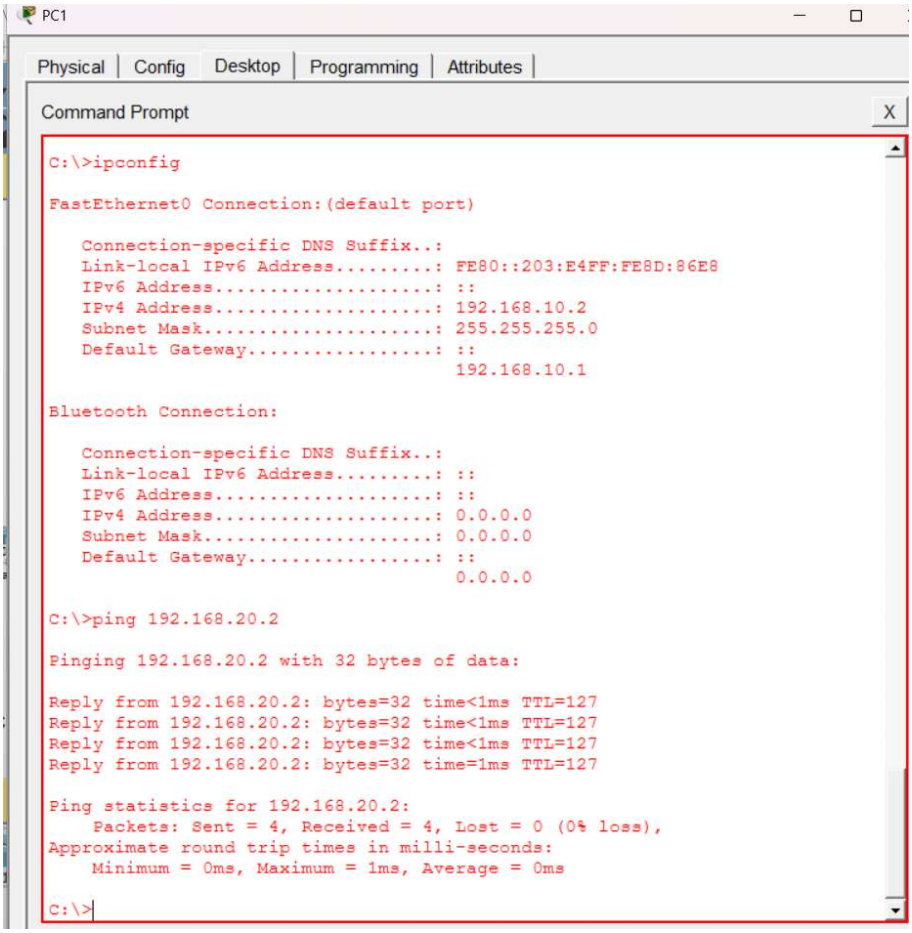
Outputs







It was determined the default-gateway for PC1 was entered wrong as 192.168.10.5 instead of 192.168.10.1 (gig0/0.10)

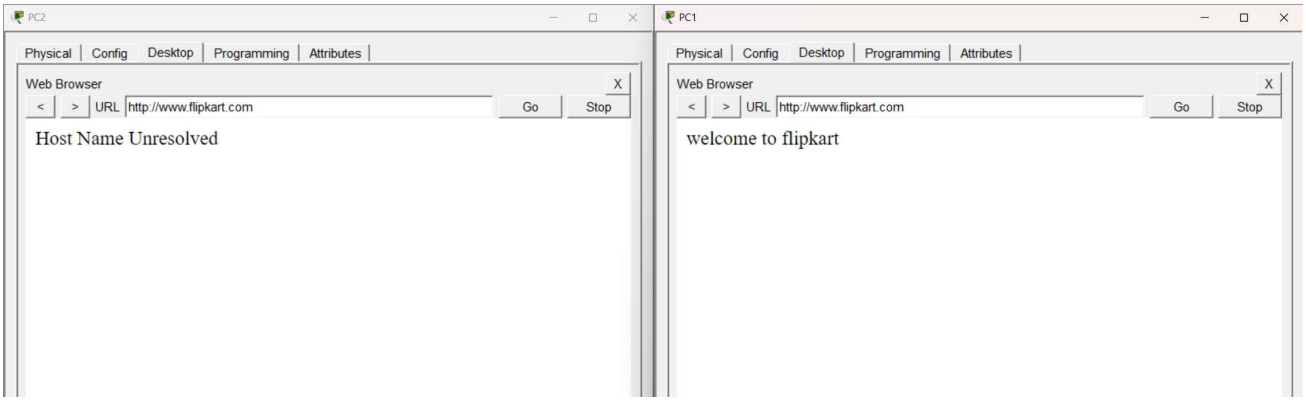


A PC sends any address outside its own subnet to its default gateway. With a wrong gateway nobody answers the ARP request, so the PC can talk to neighbors but not beyond the subnet.

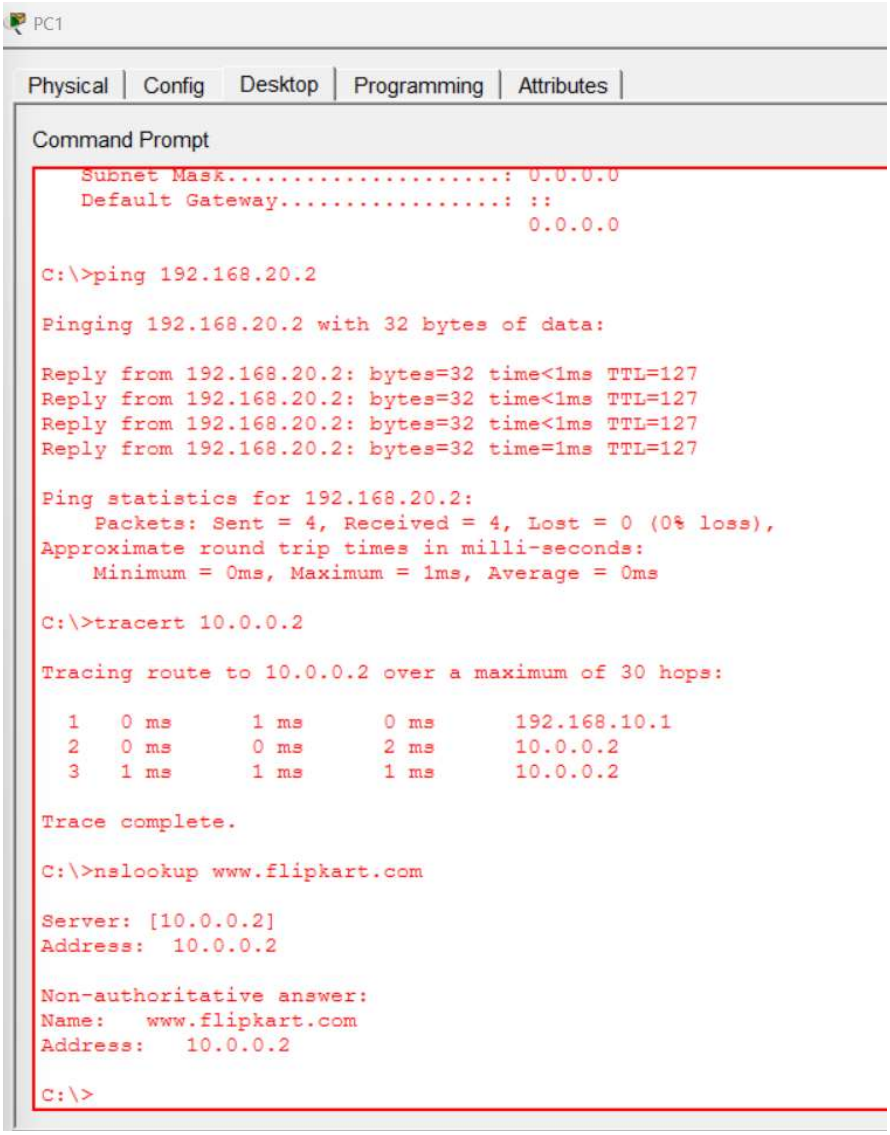
Task 3: A user reports they cannot access Flipkart.com from their simulated network. Trace the path using traceroute (or tracert) and identify where the connection fails; explain how you diagnosed and resolved the issue.

Setup

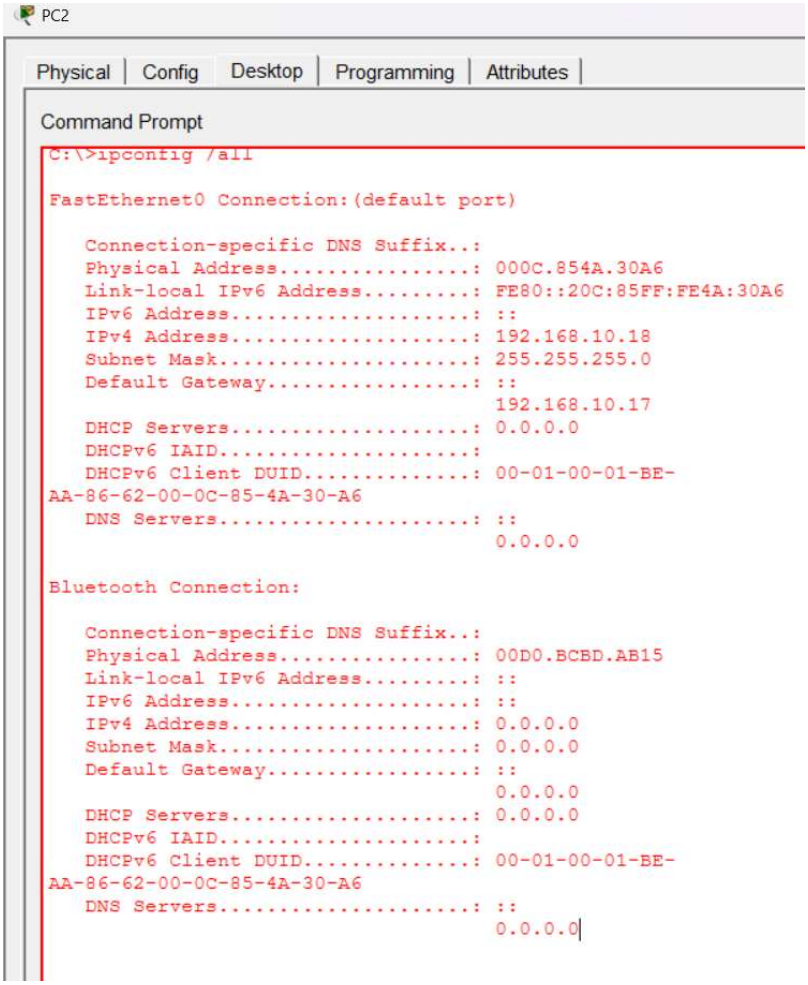
The lab server acts as an external website. In the server Services > DNS tab, a record flipkart.com = 10.0.0.2 was added, and PC1 uses 10.0.0.2 as its DNS server. The name is therefore resolved through the same NAT and routing path being tested.



Method

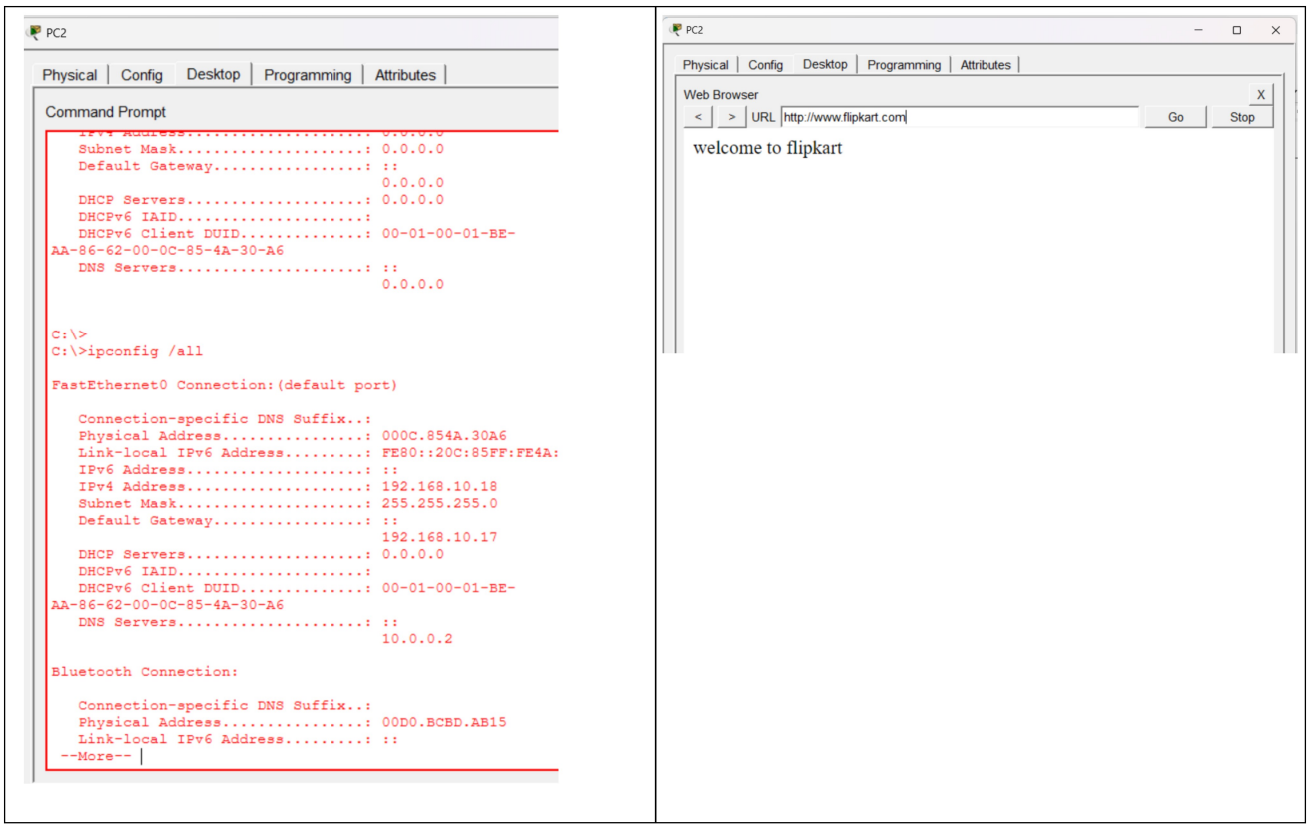


- "*Nslookup*" for flipkart.com was run in PC1 to find out its ip address, which came back as 10.0.0.2
- *"Tracert 10.0.0.2"* was run in PC1 for flipkart.com ip address and route was found.
- Same "*tracert 10.0.0.2"* was run in PC2 for flipkart.com (10.0.0.2), route was found successfully.
- But "*nslookup"* failed in PC2, this was followed with an *"ipconfig /all"* command to determine if DNS server ip address is configured.



DNS server is not configured in PC2

DNS server ip address 10.0.0.2 was configured in PC2 and then www.flipkart.com was tried in PC2 succeed.



Task 4: Simulate a scenario where a router has an incorrect static route that breaks connectivity between two VLANs. Correct the route configuration and verify the solution using appropriate show commands.

Routing mistakes found and fixed in the lab

Two real routing mistakes were made while building the lab. The first was a static route typed with the wrong mask: the host 10.0.0.2 cannot use a /24 mask because it has host bits set, so IOS rejected it with "Inconsistent address and mask". It was re-entered as a /32 host route.

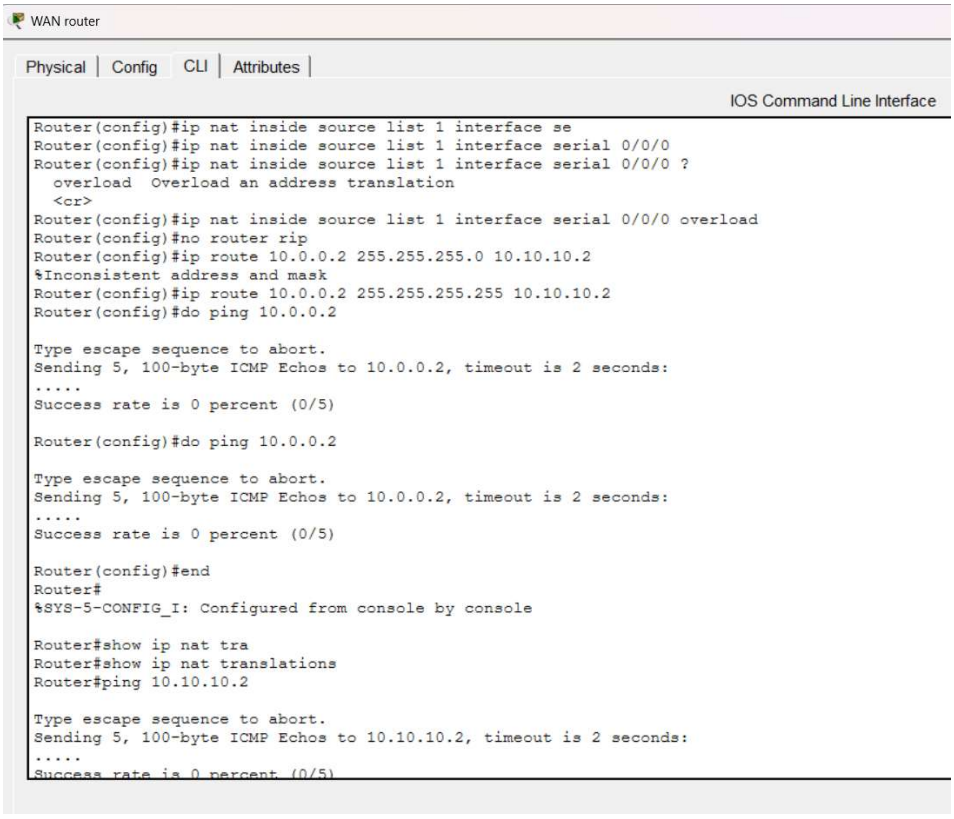


Figure 5: WAN router: route rejected with the /24 mask, then accepted as 255.255.255.255

The second was on the ISP router: Serial0/0/0 had the mask 255.0.0.0, so show ip route listed the whole 10.0.0.0/8 as directly connected. That made the server's NAT address 10.0.0.2 look like it was part of the serial link.

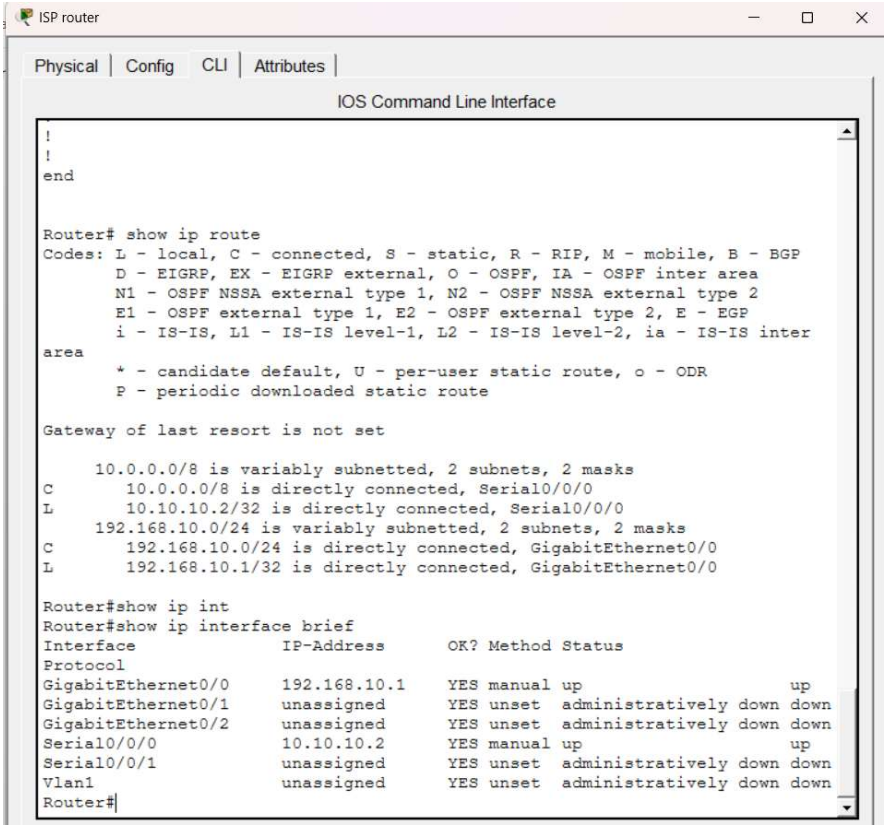


Figure 6: ISP router before the fix: C 10.0.0.0/8 on Serial0/0/0 (wrong mask)

After setting 255.255.255.252 the connected route became 10.10.10.0/30, and the WAN router holds the host route to 10.0.0.2 through 10.10.10.2.

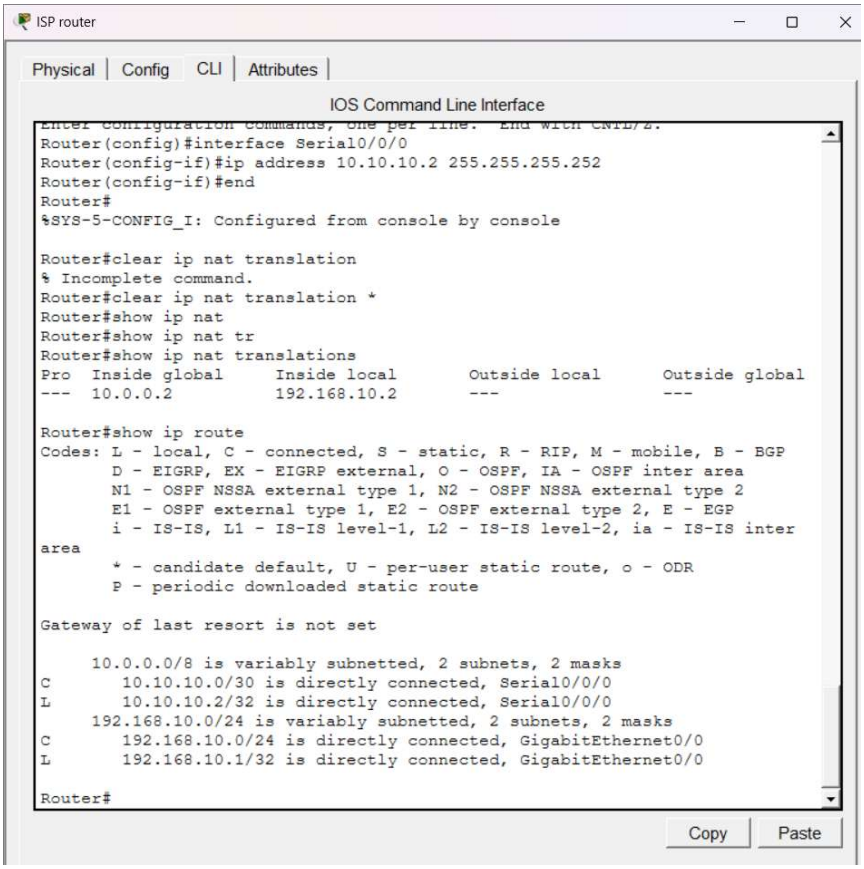


Figure 7: ISP router after the fix: C 10.10.10.0/30 on Serial0/0/0

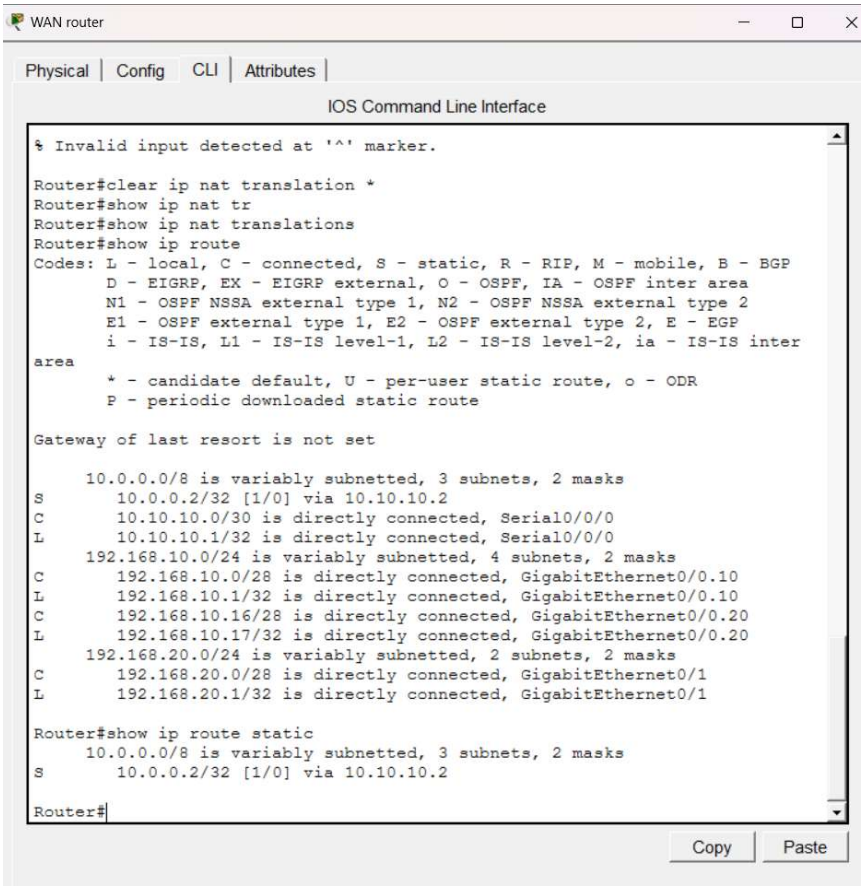


Figure 8: WAN router: show ip route and show ip route static, with S 10.0.0.2/32 via 10.10.10.2

Task 5: You receive a network diagram and a set of running-config outputs for a multi-router setup where WhatsApp voice calls are not working between two branches. Analyze the configs, identify the misconfiguration, and describe your troubleshooting steps and resolution.

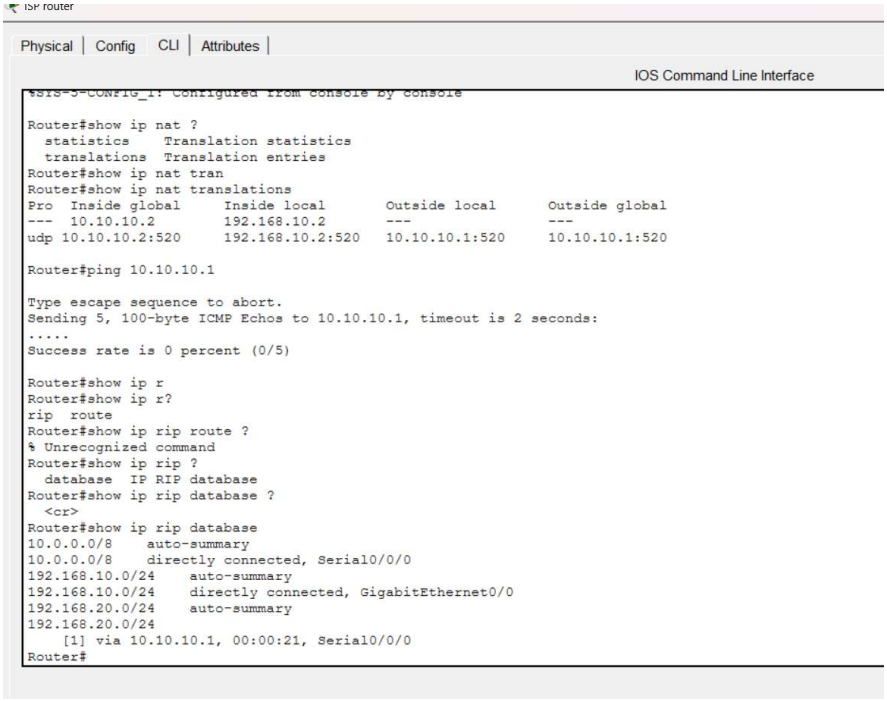
Scenario

This lab is used as the worked example: two routers (WAN router and ISP router) with NAT at each end, and a branch (the server side) that cannot be reached from the main LAN. Voice calls such as WhatsApp typically need traffic in both directions (UDP) through NAT. If either direction is untranslated or has no route, one-way tests may pass but calls fail. The symptom here is the same: PCs reach the ISP router but not the server behind it.

Troubleshooting steps

Step 1: From the PCs, 10.10.10.2 answers but 10.0.0.2 does not (Figures 2 and 3), so the LAN, VLANs, gateways and serial link are fine and the fault lies beyond the ISP router.

Step 2: The first design used 10.10.10.2 as the server's NAT address, which is the ISP router's own serial IP. The NAT table showed 10.10.10.2 mapped to 192.168.10.2, and a ping from the ISP router to 10.10.10.1 failed 0/5, because the router treated traffic meant for itself as traffic for the server.



ISP router: NAT entry using its own serial IP, failed ping to 10.10.10.1, and RIP database

The fix was to use a different address for the server (10.0.0.2), add the host route on the WAN router, and remove RIP, which was advertising 192.168.10.0/24 from the server side into the WAN router and clashing with VLAN 10.

Step 3: The fix included changing NAT for server side from 10.10.10.2 to 10.0.0.2 and changing the subnet for serial link between WAN router to ISP router from 10.0.0.0/8 to 10.10.10.0/32 that was 10.0.0.2 does not fall in the same network as the serial links.

ISP NAT table showed ICMP entries 10.0.0.2 to 192.168.10.2 with outside address 10.10.10.1, so requests reach the server. The old 10.10.10.2 static entry was still listed and was removed.

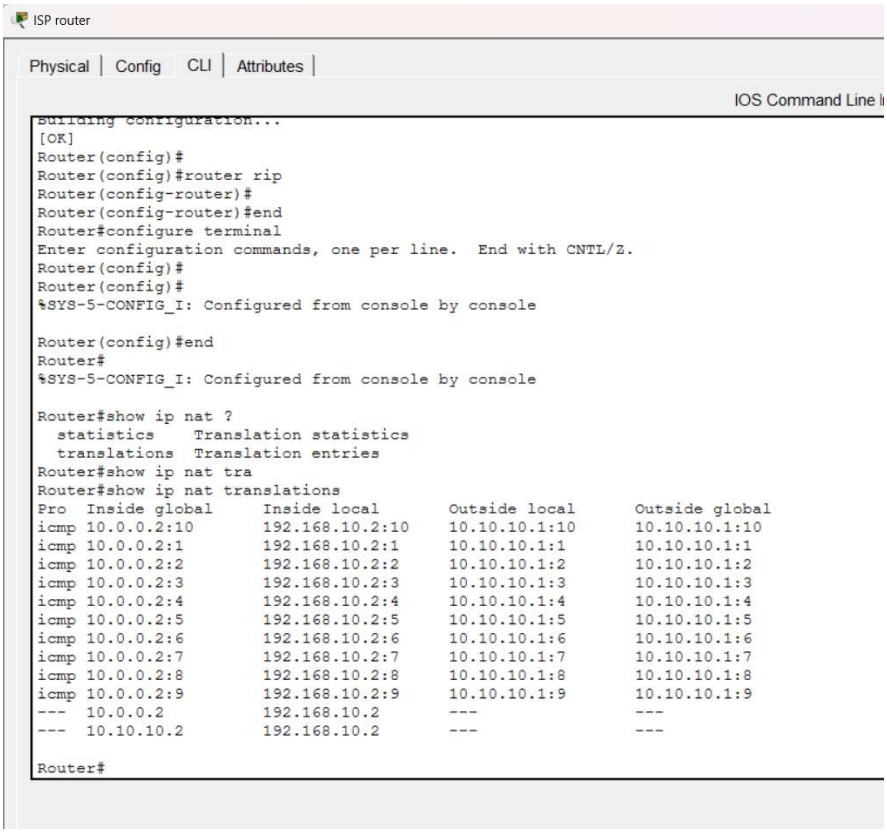


Figure 10: ISP router NAT table: requests translated to the server; old static entry still present

Step 4: The reply packet arrived at the WAN router with source 192.168.10.2 and destination 10.10.10.1. The source should have been rewritten to 10.0.0.2 by the ISP router. The WAN router found no NAT entry for it and dropped it.

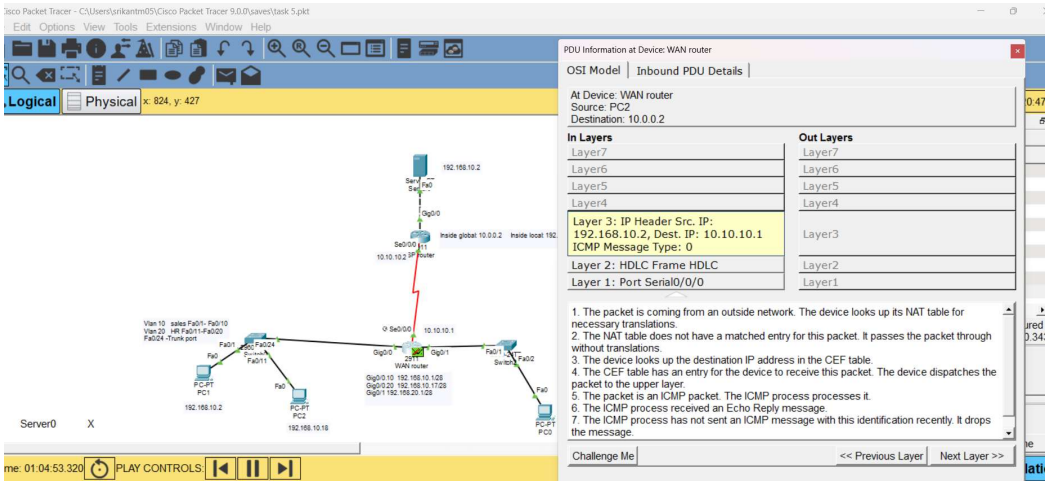


Figure 11: Simulation mode at the WAN router: reply arrives still showing the server's private address

Step 5: Compare the NAT interface roles. show running-config | include nat on the ISP router showed only one ip nat outside line and the static rule. There was no ip nat inside. NAT only translates when a packet crosses between an inside and an outside interface, so the reply was never translated.

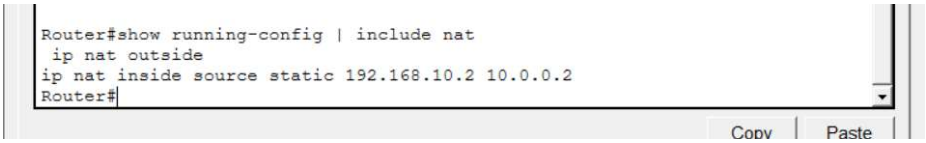
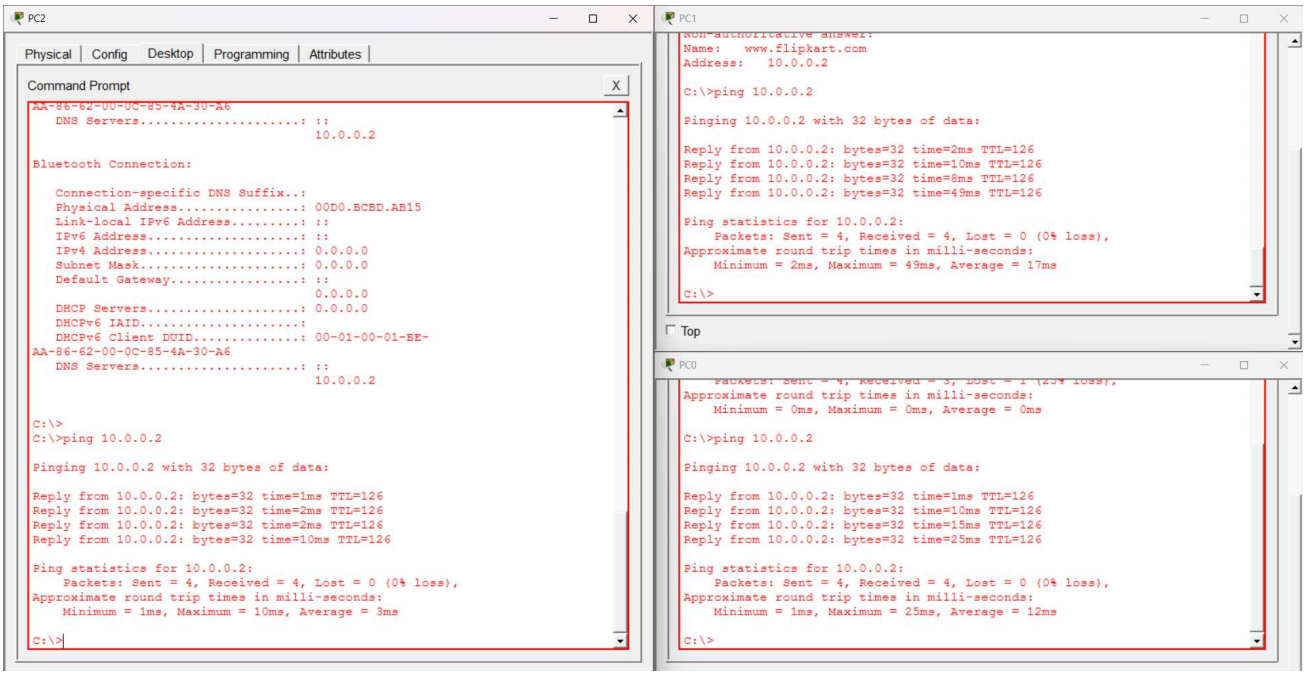


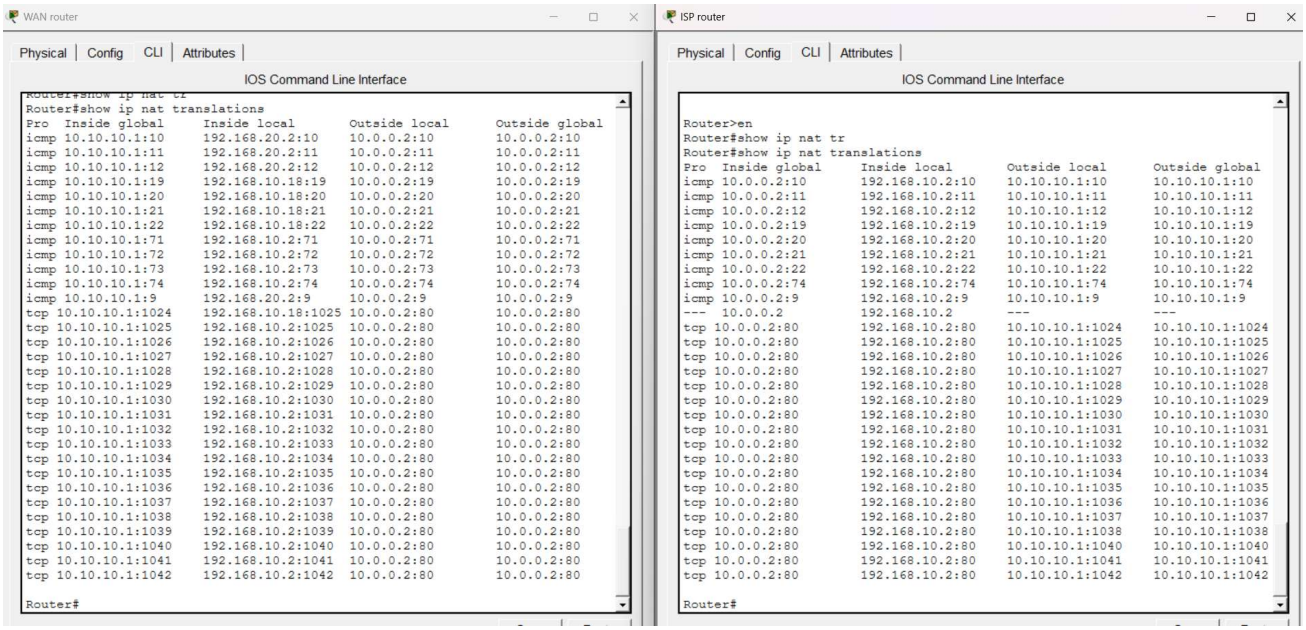
Figure 12: ISP router: ip nat outside present, ip nat inside missing

```
interface GigabitEthernet0/0
```

```
ip nat inside
```

Step 7: Verify. Clear the NAT tables, retest from every PC and the browser, and read the tables on both routers.





Summary of misconfigurations

Fault	Device	Symptom	Fix
NAT address equals the router's own serial IP	ISP	Ping to 10.10.10.1 fails; NAT table maps 10.10.10.2	Use a different inside global (10.0.0.2)
Host route typed with /24 mask	WAN	% Inconsistent address and mask	ip route 10.0.0.2 255.255.255.255 10.10.10.2

Fault	Device	Symptom	Fix
Serial mask 255.0.0.0	ISP	show ip route lists C 10.0.0.0/8	ip address 10.10.10.2 255.255.255.252
Missing ip nat inside on Gig0/0	ISP	Reply returns untranslated (Simulation mode)	Add ip nat inside on Gig0/0
RIP advertising overlapping 192.168.10.0/24	ISP/WAN	Conflicts with VLAN 10	no router rip; use static routes

Final NAT configuration

ISP router

```
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
interface Serial0/0/0
 ip address 10.10.10.2 255.255.255.252
 ip nat outside
ip nat inside source static 192.168.10.2 10.0.0.2
```

WAN router (NAT-related lines)

```
interface GigabitEthernet0/0.10
 ip nat inside
interface GigabitEthernet0/0.20
 ip nat inside
interface GigabitEthernet0/1
 ip nat inside
interface Serial0/0/0
 ip address 10.10.10.1 255.255.255.252
 ip nat outside
access-list 1 permit 192.168.10.0 0.0.0.31
access-list 1 permit 192.168.20.0 0.0.0.15
ip nat inside source list 1 interface Serial0/0/0 overload
ip route 10.0.0.2 255.255.255.255 10.10.10.2
```